

Scientists use PlayStations to create supercomputer

Computer hobbyists and researchers take note: two U.S. scientists have created a step-by-step guide on how to build a supercomputer using multiple PlayStation 3 video-game consoles.

The instructional guide, posted this week online at ps3cluster.org, allows users with some programming knowledge to install a version of the open-source operating system Linux on the video consoles and connect a number of consoles into a computing cluster or grid.

The two researchers say the guide could provide scientists with another, cheaper alternative to renting time on supercomputers to run their simulations.

University of Massachusetts Dartmouth physics professor Gaurav Khanna first built the cluster a year ago to run his simulations estimating the gravitational waves produced when two black holes merged.

Frustrated with the cost of renting time on supercomputers, which he said can cost as much as \$5,000 to run a 5,000-hour simulation, Khanna decided to set up his own computer cluster using PS3s, which had both a powerful processor developed by Sony, IBM and Toshiba, but also an open platform that allows different system software to run on it. PlayStation 3 systems retail for about \$400 Cdn.

On the how-to-guide Khanna says the eight-console cluster is roughly comparable in speed to a 200 node IBM Blue Gene supercomputer. Khanna says his research now runs using a cluster of 16 PS3s.

The fastest supercomputer in the world, IBM's Roadrunner supercomputer at Los Alamos National Laboratory, has 3,250 nodes and is capable of 1.105 petaflops, or 1.105 quadrillion floating point operations per second, about 100,000 times faster than a home computer.

Massachusetts Dartmouth computer scientist Chris Poulin, who co-wrote the instructional manual with Khanna, wouldn't reveal the number of flops the system can achieve, but said anecdotally the cluster has allowed him to run simulations in hours that used to take days on a powerful server computer.

Khanna's not the first researcher to use PS3s to simulate the effects of a supercomputer. The University of Stanford's Folding at Home project allows people to help with research into how proteins self-assemble — or fold — by downloading software onto their home PS3s, creating a virtual supercomputer. Their research is currently targeting proteins relevant to diseases such as Alzheimer's and Huntington's disease.

But the guide posted by Khanna and Poulin is the first that might allow someone to set up a supercomputer in their own home.

Poulin said there are two major practical issues, however, that might limit the practicality of a PS3 cluster supercomputer.

The first issue is power. He said the video-game consoles use about 200 to 300 watts per unit, so finding a room that could hook up eight of the consoles might be an issue for hobbyists, he says.

"I think if you put four or more than four of the systems on one plug you'd probably blow a fuse," Poulin told CBC News.

The second issue is memory. The console has only 256 MB of RAM, far less than most personal computers available now. Poulin said that while the low memory wouldn't be a problem for straightforward computations, running multiple simulations or programs could tax the system. As a result, simulations running on the cluster would have to be tailored to consider the cluster's memory limitations.

Poulin said he hopes the project will help open doors to more partnerships between industry and universities that will lead to better access to supercomputing power.

"That's ultimately the goal here," he said. "We want to make things easier, no matter what kind of supercomputer you are using."